Synoptic Question 30: The Physics of an Automatic Washing Machine

Reading Passage

Read the following article and then answer the questions that follow.

An automatic washing machine uses a combination of mechanical action, thermal energy, and chemical agents (detergents) to clean clothes. A key component is a rotating drum, usually mounted horizontally or slightly tilted. Clothes are placed inside the drum, which is then partially filled with water.

An electric motor drives the drum's rotation. During the wash cycle, the drum rotates back and forth at a relatively low speed, tumbling the clothes through the water. For the spin cycle, the drum rotates at a very high speed. The resulting centripetal force pushes the clothes against the perforated drum wall, while the water passes through the holes and is pumped out, effectively drying the clothes.

Many washing machines also include an electrical heating element submerged in the water. This element, essentially a large resistor, converts electrical energy into thermal energy to heat the water to a desired temperature, which improves the effectiveness of the detergent. The power consumption of the machine is dominated by the heating element and the motor, particularly during the high-speed spin cycle.

Modern machines often use electronic controls to manage the cycle timing, water level, temperature, and motor speed. The motor itself is typically an AC induction motor or a brushless DC motor, chosen for its efficiency and ability to provide variable speeds.

Data:

Data for a typical front-loading washing machine:

Table 1: Data for Washing Machine Analysis

Parameter	**Value**
Inner radius of drum, r	0.25 m
Length of drum, L	0.30 m
Mass of empty drum, m_{drum}	8.0 kg
Maximum mass of dry clothes, m_{load}	7.0 kg
Volume of water used in wash cycle	50 litres
Density of water, ρ_{water}	1000 kg m^{-3}
Specific heat capacity of water, c_{water}	4200 J kg^{-1} K^{-1}
Rotational speed during wash cycle	50 rpm
Rotational speed during spin cycle, $\omega_{spin,rpm}$	1200 rpm
Time taken to accelerate to spin speed, t_{accel}	4.0 s
Power rating of heating element, P_{heat}	2100 W
Mains voltage supply (RMS), V_{rms}	230 V
Efficiency of motor during spin, η_{motor}	75%

Questions

(a) During the wash cycle, water is heated.

(i) Calculate the mass of water used in the wash cycle. (1 litre = 1×10^{-3} m^3).

(ii) Calculate the energy required to heat this water from 15.0 °C to 40.0 °C.

(iii) Assuming the heating element is the only component drawing power, calculate the time taken to heat the water.

(iv) Calculate the resistance of the heating element.

(b) During the spin cycle, the drum rotates at high speed. Assume the clothes (m_{load}) stick uniformly to the inner wall of the drum.

(i) Convert the spin speed of 1200 rpm into angular velocity ω in rad s^{-1}.

(ii) Calculate the linear speed v of the clothes during the spin cycle.

(iii) Calculate the magnitude of the centripetal force acting on a small 0.10 kg piece of clothing during the spin.

(iv) Explain qualitatively why the water is extracted from the clothes during the spin cycle.

(c) Consider the acceleration phase of the spin cycle. The total moment of inertia I_{total} of the drum and wet clothes can be approximated as $I_{total} \approx (m_{drum} + m_{wet_load})r^2$. Assume the wet load mass is double the dry load mass.

(i) Estimate the moment of inertia I_{total} of the loaded drum.

(ii) Assuming constant angular acceleration, calculate the angular acceleration α as the drum speeds up.

(iii) Calculate the net torque τ_{net} required to produce this acceleration.

(iv) Calculate the work done W by this net torque during the acceleration phase.

(d) The motor provides the torque to accelerate and spin the drum.

(i) Calculate the maximum rotational kinetic energy of the loaded drum during the spin cycle.

(ii) Assuming the work done in (c)(iv) equals the final rotational KE, calculate the average mechanical power P_{mech} supplied by the motor during acceleration.

(iii) Calculate the average electrical power P_{elec} consumed by the motor during acceleration.

(e) Briefly explain the basic principle (interaction between magnetic fields and currents) by which an electric motor produces torque.

Solutions

(a) (i) Volume $V = 50$ litres $= 50 \times 10^{-3}$ m^3.
Mass $m_{water} = \rho_{water} V = (1000)(50 \times 10^{-3}) = 50.0$ kg.

(ii) $\Delta T = 40.0 - 15.0 = 25.0$ K.
Energy $Q = m_{water} c_{water} \Delta T = (50.0)(4200)(25.0) = 5.25 \times 10^6$ J.

(iii) Power $P_{heat} = 2100$ W.
Time $t = \frac{Q}{P_{heat}} = \frac{5.25 \times 10^6}{2100} = 2500$ s (approx 41.7 minutes).

(iv) $P = \frac{V_{rms}^2}{R} \implies R = \frac{V_{rms}^2}{P_{heat}}$
$R = \frac{(230)^2}{2100} = 25.19\ \Omega \approx 25.2\ \Omega$.

(b) (i) $\omega = 1200 \frac{\text{rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 125.7$ rad s^{-1}.

(ii) $v = r\omega = (0.25)(125.7) = 31.43$ m s$^{-1} \approx 31.4$ m s^{-1}.

(iii) $m_{piece} = 0.10$ kg.
$F_c = m_{piece} r \omega^2 = (0.10)(0.25)(125.7)^2 = 395$ N.
(Alternatively: $F_c = m_{piece} v^2 / r = (0.10)(31.43)^2/0.25 = 395$ N).

(iv) The drum wall provides the large inward centripetal force required to keep the clothes moving in a circle. Water droplets, being mobile, are not held as strongly by the fabric. Their inertia means they tend to continue moving in a straight line (tangential to the circle). They pass through the perforations in the drum and are collected and pumped away.

(c) $m_{wet_load} = 2 \times m_{load} = 2 \times 7.0 = 14.0$ kg.

(i) $I_{total} \approx (m_{drum} + m_{wet_load}) r^2 = (8.0 + 14.0)(0.25)^2 = (22.0)(0.0625) = 1.375$ kg m^2.

(ii) $\omega_f = 125.7$ rad s^{-1}, $\omega_i = 0$.
$\alpha = \frac{\omega_f - \omega_i}{t_{accel}} = \frac{125.7 - 0}{4.0} = 31.4$ rad s^{-2}.

(iii) $\tau_{net} = I_{total} \alpha = (1.375)(31.4) = 43.175$ Nm ≈ 43.2 Nm.

(iv) Angular displacement $\theta = \omega_i t + \frac{1}{2}\alpha t^2 = 0 + \frac{1}{2}(31.4)(4.0)^2 = 251.2$ rad.
Work done $W = \tau_{net}\theta = (43.175)(251.2) = 10845$ J $\approx 1.08 \times 10^4$ J.
(Alternatively: $W = \Delta KE_{rot} = \frac{1}{2} I \omega_f^2 = \frac{1}{2}(1.375)(125.7)^2 = 10857$ J). Let's use 10.9 kJ.

(d) (i) $KE_{rot,max} = \frac{1}{2} I_{total} \omega_{spin}^2 = \frac{1}{2}(1.375)(125.7)^2 = 10857$ J ≈ 10.9 kJ.

(ii) $P_{mech,avg} = \frac{W}{t_{accel}} = \frac{10857}{4.0} = 2714$ W ≈ 2.71 kW.

(iii) $\eta_{motor} = \frac{P_{mech}}{P_{elec}} \implies P_{elec} = \frac{P_{mech}}{\eta_{motor}}$
$P_{elec,avg} = \frac{2714}{0.75} = 3619$ W ≈ 3.62 kW.

(e) An electric motor typically uses coils carrying an electric current placed within a magnetic field. The magnetic field exerts a force (the Lorentz force, $F = BIL \sin\theta$) on the current-carrying wires. These forces create a turning effect, or torque, on the coil assembly (rotor), causing it to rotate and drive the drum.

Cambridge International AS & A Level Physics (9702)

Paper 4 Structured Questions

Variant 2 (Higher Difficulty): Washing Machine Physics

Answer all questions.
Give all working and appropriate units.
Total: 50 marks

Comprehension Passage

A front-loading washing machine is tested in a lab. During the main wash, the machine uses 42 L of water and raises its temperature from 12 °C to 55 °C. The heating element is rated at 2.20 kW on a 230 V RMS supply.

In the spin cycle, the drum accelerates from rest to 1400 rpm in 5.0 s. At the start of spin, the wet laundry mass is found to be 1.85 times the dry laundry mass. During controlled braking, the drum decelerates to rest in 6.0 s, and the regenerative system returns energy with an efficiency of 0.60.

Use this passage and the data table below to extract relevant information for the questions.

Data

Parameter	Value
Drum inner radius, r	0.26 m
Drum length, L	0.32 m
Empty drum mass, m_{drum}	8.5 kg
Dry clothes mass, m_{dry}	8.0 kg
Specific heat capacity of water, c	4200 $\text{J kg}^{-1}\text{K}^{-1}$
Motor mechanical efficiency (motoring), η_m	0.82

Take density of water as $\rho = 1000\,\text{kg m}^{-3}$ and $g = 9.81\,\text{m s}^{-2}$.

1

1.1 Determine the mass of water heated in the wash cycle. [2]

1.2 Calculate the thermal energy required to heat the water from 12 °C to 55 °C. [3]

1.3 The measured heating time is 48 min. Estimate the overall heating efficiency,

$$\eta_h = \frac{\text{useful thermal output}}{\text{electrical input}}.$$

Comment on your result. [3]

1.4 Determine the resistance of the heating element at rated voltage. [2]

2

2.1 Convert 1400 rpm to angular velocity in rad s^{-1}. [2]

2.2 Determine the linear speed at the drum wall. [2]

2.3 Calculate the centripetal acceleration at the drum wall, and express your answer in units of g. [3]

2.4 A 0.12 kg section of fabric is at the drum wall. Calculate the centripetal force on this section. [3]

3

At the start of spin, assume

$$m_{\text{wet}} = m_{\text{dry}}(1 + 0.85)$$

and model the drum + wet load as a thin cylindrical shell:

$$I \approx (m_{\text{drum}} + m_{\text{wet}})r^2.$$

3.1 Calculate I. [2]

3.2 Calculate the angular acceleration during spin-up. [2]

3.3 Determine the net torque required during spin-up. [2]

3.4 Determine the rotational kinetic energy at full spin speed. [3]

3.5 Determine the average mechanical power delivered to the drum during spin-up. [1]

3.6 Estimate the average electrical input power to the motor during spin-up. [2]

4

4.1 Assuming all rotational kinetic energy at full speed is available to braking, determine the ideal recoverable energy. [2]

4.2 Using $\eta_r = 0.60$, estimate the electrical energy returned during braking. [2]

4.3 Determine the average regenerated electrical power over 6.0 s deceleration. [2]

5

The statement below is made:

> "Water leaves the clothes in the spin cycle because the centrifugal force throws it outward."

Evaluate this statement using an inertial-frame explanation. You may also refer to rotating-frame ideas used in engineering. [6]

6

A manufacturer wants to increase spin speed from 1400 rpm to 1600 rpm, while keeping peak electrical input power unchanged.

Propose and justify **two** engineering modifications. For each modification, include one trade-off. [6]

Cambridge International AS & A Level Physics (9702)
Paper 4 Mark Scheme
Variant 2: Washing Machine Physics

Total: 50 marks

1 (10 marks)

1.1 $m = \rho V = (1000)(42 \times 10^{-3}) = 42.0\,\text{kg}$ (B1), correct substitution and value (A1). [2]

1.2 $Q = mc\Delta T = (42)(4200)(43) = 7.59 \times 10^6\,\text{J}$ (M1A1A1). [3]

1.3 $E_{\text{in}} = P_h t = (2200)(48 \times 60) = 6.34 \times 10^6\,\text{J}$ (M1), $\eta_h = Q/E_{\text{in}} \approx 1.20$ or 120% (A1), states non-physical/inconsistent data or assumptions (B1). [3]

1.4 $R = V^2/P = 230^2/2200 = 24.0\,\Omega$ (M1A1). [2]

2 (10 marks)

2.1 $\omega = 1400 \times 2\pi/60 = 146.6\,\text{rads}^{-1}$ (M1A1). [2]

2.2 $v = r\omega = (0.26)(146.6) = 38.1\,\text{ms}^{-1}$ (M1A1). [2]

2.3 $a_c = r\omega^2 = 5.59 \times 10^3\,\text{ms}^{-2}$ (M1A1), $a_c/g \approx 570$ (A1). [3]

2.4 $F = ma = (0.12)(5.59 \times 10^3) = 671\,\text{N}$ (M1A1A1). [3]

3 (12 marks)

3.1 $m_{\text{wet}} = 8.0(1.85) = 14.8\,\text{kg}$, $I = (8.5 + 14.8)(0.26)^2 = 1.58\,\text{kgm}^2$ (M1A1). [2]

3.2 $\alpha = \Delta\omega/\Delta t = 146.6/5.0 = 29.3\,\text{rads}^{-2}$ (M1A1). [2]

3.3 $\tau = I\alpha = (1.58)(29.3) = 46.2\,\text{Nm}$ (M1A1). [2]

3.4 $E_k = \frac{1}{2}I\omega^2 = 1.70 \times 10^4\,\text{J}$ (M1A1A1). [3]

3.5 $\bar{P}_{\text{mech}} = E_k/t = 3.40\,\text{kW}$ (C1). [1]

3.6 $\bar{P}_{\text{elec}} = \bar{P}_{\text{mech}}/\eta_m = 3.40/0.82 = 4.15\,\text{kW}$ (M1A1). [2]

4 (6 marks)

4.1 $E_{\text{ideal}} = E_k = 17.0\,\text{kJ}$ (B1B1). [2]

4.2 $E_{\text{return}} = \eta_r E_{\text{ideal}} = 0.60 \times 17.0 = 10.2\,\text{kJ}$ (M1A1). [2]

4.3 $\bar{P}_{\text{regen}} = E_{\text{return}}/6.0 = 1.70\,\text{kW}$ (M1A1). [2]

5 (6 marks)

Indicative content (credit any coherent equivalent physics):

- inertial-frame account: inward force required for circular motion of cloth (B1)
- water less constrained than fabric; tends to continue tangentially when not fully constrained (B1)
- separation and passage through perforations linked to inertia and contact constraints (B1)
- states centrifugal force is pseudo-force in rotating frame (B1)
- correctly uses rotating-frame language as modelling tool (B1)

- logical, technically consistent explanation with clear terminology (QWC/B1)

6 (6 marks)
Award up to 3 marks per modification: valid physics-based proposal (1), clear justification (1), explicit trade-off (1). Examples include: longer controlled spin-up profile, improved balancing/sensing, reduced rotational inertia, drum geometry optimisation, higher stiffness/suspension redesign, advanced inverter control.
General principle: allow ecf and method marks where physics approach is valid and units are consistent.

Model Answers (for teacher use)

Q5 sample response: In an inertial frame, the cloth is forced into circular motion by inward contact forces from the drum. Water in the cloth is less strongly constrained, so some water does not follow the required curved path and instead continues approximately tangentially due to inertia. Relative to the rotating drum, this appears as outward movement, and water reaches drum perforations and exits. Thus, "centrifugal force throws water out" is an acceptable rotating-frame shorthand, but fundamentally the separation is explained by inertia and differing constraints, not by a real outward interaction force.

Q6 sample response: (i) Increase spin-up time via inverter control so peak power remains capped while final speed reaches 1600 rpm; trade-off is longer cycle time. (ii) Improve dynamic balancing (better sensing and redistribution pauses), reducing vibration losses so more available power increases useful rotational energy; trade-off is greater control complexity and cost.